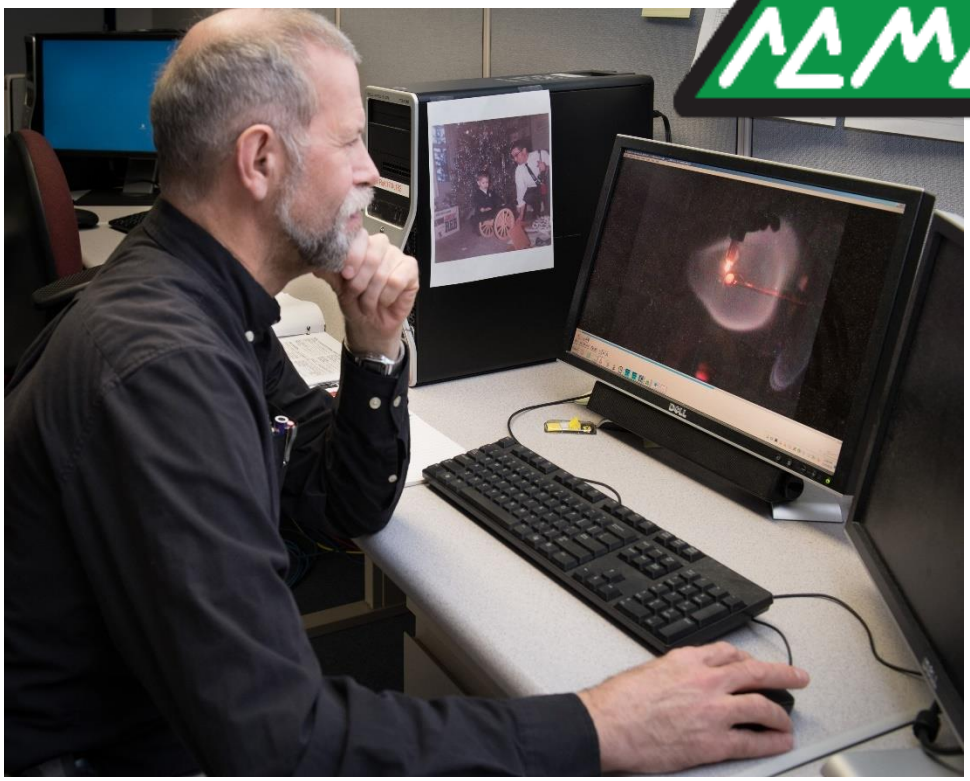
Advanced Combustion via Microgravity Experiments (ACME)
Structure and Response of Spherical Diffusion Flames (s-Flame)

ACME/s-Flame Data Interpretation

Prepared by:
Dennis P. Stocker*, NASA Glenn Research Center*
dennis.p.stocker@nasa.gov, 216-433-2166



Dennis Stocker, the ACME Project Scientist, at the science console for the Combustion Integrated Rack (CIR) in NASA's Glenn ISS Payload Operations Center (GIPOC). In this photo, he is reviewing data from an s-Flame test.

Commercial products are identified for reference and are not meant to indicate endorsement by NASA or the federal government.

Table of Contents

	Page
1. Introduction	**2**
2. s-Flame Overview	**3**
3. Tests Conducted	**3**
4. Data Interpretation	**5**
A. Non-Imaging Data	**6**
1. Data Output File	6
2. Mass Flow Controller (MFC) Setpoint File	8
3. Gas Chromatograph Instrumentation Package (GCIP) data	8
4. Space Acceleration Measurement System (SAMS) data	9
B. Image Files	**10**
1. Analog, operations (ops), internal camera	10
2. ACME Data Camera (DataCam)	10
3. Cool Flames Investigation (CFI) intensified camera	15
4. Low Light Level Ultraviolet (LLL-UV) camera	18
5. High Bit-depth Multi-spectral (HiBMs) camera and CIR illumination package	18
C. Test Parameters and Other Files	**18**
1. CIR chamber calculator file	19
2. Test parameter file	20
3. Notes	20
4. Test timing	21
5. Logs	21
Appendices	**31**
A. Nomenclature	**31**
B. Installation of QtOMA2 on a PC	**32**

1. Introduction

The purpose of this report is to describe the data acquired for ACME's s-Flame experiment to facilitate its analysis. The data is all digital, where the fuel was gaseous and there was no collection of soot or the combustion gases. For each test, the key data consists of image files and a file with the non-imaging measurements. Additional files such as chamber calculator output, camera metadata, and operational logs provide information that can support the analysis.

It is recommended that readers interested in analyzing the s-Flame data begin by familiarizing themselves with the hardware and instrumentation, either through the overview in the *s-Flame Final Report* or the more comprehensive description found in the *ACME Engineering Implementation* document.

Those seeking to analyze the data should next review this document. To interpret the non-imaging data, it is suggested that a sample data file be opened, named for example 19254A1_data.csv, which has comma-separated values and can be opened with spreadsheet software such as Microsoft Excel. The data is in raw voltages, where conversion to engineering units requires study of ACME's Software Interface Design Description, i.e., ACME-DOC-010 (where the latest version at the time of this writing is revision D). Additional documents that may be important include:

ACME-RPT-010 A	Data Camera Characterization Report, Rev. A
ACME-RPT-011	Radiometer Characterization Report
ACME-RPT-013	PMT Characterization Report
MDC-XXX-1866-R1	CFI Intensified Camera Package Characterization Report

Meanwhile, the BRE and Cool Flames radiometer characterization reports are **not applicable to s-Flame** because it used the standard radiometers addressed in ACME-RPT-011.

2. s-Flame Overview

This text is the "Overview" from the ACME/s-Flame Final Report, which provides descriptions of the science, preliminary results, and more. It is suggested that it be reviewed by those interested in analyzing the s-Flame data.

The Structure and Response of Spherical Diffusion Flames (s-Flame) experiment is one of six independent experiments that make up the Advanced Combustion via Microgravity Experiments (ACME) project for which tests are conducted within the Combustion Integrated Rack (CIR) on the International Space Station (ISS).

The experiment's Principal Investigator (PI) is Prof. Chung K. Law of Princeton University. The experiment has two Co-Investigators (Co-Is), namely Prof. Stephen D. Tse of Rutgers University and Dr. Kurt R. Sacksteder of the NASA Glenn Research Center. While Sacksteder was actively engaged in defining the experiment, he had transitioned to other NASA work long before the spaceflight operations limiting his overall contributions.

Meanwhile, Dennis P. Stocker of the NASA Glenn Research Center is the Project Scientist (PS) for ACME including s-Flame and the rest of the ACME investigations. In preparatory drop testing, etc., Dr. Fumiaki Takahashi of the Case Western Reserve University (CWRU) served as the Deputy Project Scientist. For ACME's spaceflight operations, the NASA science team includes Drs. Vedha Nayagam and Uday G. Hegde who are also with CWRU.

The s-Flames operations were conducted in two rounds of testing during Increments 60-63, where there was no testing during Increment 62 due to limited crew availability. More specifically, the first round of testing was conducted from Sept. 2019 to Jan. 2020, while the second-round testing – featuring helium rather than nitrogen dilution - occurred from May to July 2020.

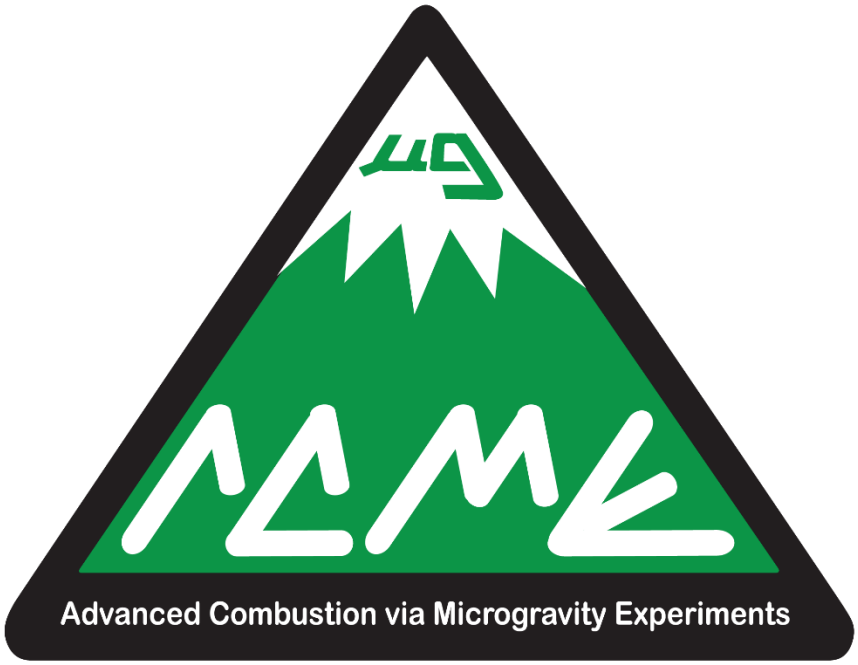
3. Tests Conducted

This text is an excerpt from the "Results" section of the ACME/s-Flame – 30-Day Operation Accomplishments Report for ISS Increment 63 (which is included in the ACME/s-Flame Final Report).

> A total of 276 flames were ignited for the s-Flame experiment, with 29, 116, and 131 respectively ignited during Increments 60, 61, and 63. These numbers neglect some flames which quenched promptly after ignition. In this regard, science testing began at the end of Increment 60 and the first round of tests were completed during Increment 61. No tests were conducted in Increment 62 due to a lack of crew time, where the second round of tests were fully conducted during Increment 63.

The tests were conducted in two rounds, where the first was conducted in increments 60 and 61 featured 145 flames (nominally) in which the fuel and oxygen (i.e., in the chamber atmosphere) were diluted with nitrogen. The second round conducted in increment 63 featured 131 flames (nominally) in which the fuel and oxygen were instead diluted with helium.
The s-Flame experiment does not have living subjects like a life science experiment, but the tests are independently conducted with various gaseous fuels, including hydrogen, methane, hydrogen/methane blends, and ethylene. These are simple hydrocarbon gases for which the chemistry is relatively well understood.

No samples (e.g., of soot) were collected for s-Flame; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.



A summary log of the flames ignited for s-Flame, including a limited overview of the test conditions, is found in Table 1.

Table 1. Flames ignited for the s-Flame experiment

Test Day[1]	Year	Month	Date	Atmosphere[2]	Fuel bottle	Flames ignited[3]
First round of tests, with nitrogen-diluted fuel and oxygen						
19254	2019	9	11	~21% O_2 with N_2	50% C_2H_4 with N_2	13
19256			13			16
19304		10	31		45% CH_4 with N_2	9
19308		11	4	~30% O_2 with N_2		21
19352		12	18	~21% O_2, with N_2	100% H_2	10
20006	2020	1	6		44.4% H_2 with CH_4	17
20013			13		66.7% H_2 with CH_4	22
20024			24		100% H_2	16
20029			29		50% C_2H_4 with N_2	21
Second round of tests, with helium-diluted fuel and oxygen						
20132	2020	5	11	~21% O2 with He	25% H2, 20% CH4, with He	7
20136			15		20% H2, 25% CH4, with He	11
20142			21	~30% O2 with He	30% H2, 15% CH4, with He	17
20155		6	3	~30% and ~21% O2 with He	45% H2 with He	9
20160			8		45% CH4 with He	12
20164			12	~30% O2 with He	20% C2H4 with He	11
20170			18	~30% and ~21% O2 with He	25% C2H4 with He	20
20176			24		30% C2H4 with He	23
20190		7	8		45% H2 with He	21

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Atmosphere** is ~1 atm but pressure increases and O2% drops with vitiation.
[3]**Flames ignited** is the number of nominally sustained flames ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

4. Data Interpretation

This key section provides (1) a general description of the raw experimental data and associated files and (2) guidance on data conversions, e.g., to engineering units. Only limited guidance is provided for the processing of image data, with the important exception of the conversion of the ACME DataCam images from the raw Halcon object (HOBJ) format.

A. Non-Imaging Data

Data and files discussed in this section include the following:

Data Output File
Mass Flow Controller (MFC) Setpoint File
Gas Chromatograph Instrumentation Package (GCIP) data
Space Acceleration Measurement System (SAMS) data

1. Data Output File

The basic ACME non-imaging measurements for each test are together in a file described in Section 8.1 Data Output File on page 124 of Software Interface Design Description (ACME-DOC-010 Rev D). The test number is made up of the year (2-digts), day of year (3 digits), and test (a letter and single-digit number). As such, the example, 19254A1 is test A1 conducted on the 254th day of 2019. But the test could also be early on the next (e.g., 255th) day because the day portion of ACME test numbers is not changed when operations continue past midnight (whether GMT or Eastern) into the start of the next day.

Tests numbers that only differ by the final digit (e.g., 19254H1, 19254H2, and 19254H3) are normally similar and occasionally identical, where the number is indicative of the variant or repeat. Letters I and O are not used because of their similarity to 1 and 0. While initially used in ACME testing, the letter E ceased to be used because Excel software (for example) by default interpreted it as representing an exponential which was inconvenient. As a related aside, ACME tests are often not conducted in order and some planned tests are regularly not conducted at all. The latter results in additional letter gaps if the tests from a day are sorted alphabetically. However, ACME testing often begins with test A1, where the final digit begins with 1 and is increased from there with variant or repeat tests for the same letter.

The basic data is in a CSV format, i.e., with comma-separated values, which can be opened by spreadsheet software such as Microsoft Excel. The first row is a header and provides a limited summary of the hardware and operational conditions. The header is explained in Section 10.1 Metadata Header of ACME-DOC-010 Rev D, which begins on page 126.

As an example, the header from 20190A1_data.csv includes the following information (displayed in two columns rather than a long row):

2020-07-08 190 13:53:32:617
Experiment: 2
Test Point: 20190A1
MFCA1 Gas: 4
MFCA1 Full Scale: 20.000000
MFCA2 Gas: 0
MFCA2 Full Scale: 0.500000
MFCA3 Gas: 14
MFCA3 Full Scale: 1.000000
EField: 0
EFieldPol: 0
Igniter: 0
Burner: 0
Thermocouple: 1
Radiometer: 0
IPSU 1: 1
IPSU 2: 1
IPSU 3: 0
IPSU 4: 0
IPSU 5: 1
IPSU 6: 0
Fiber: 1
PMT 1 Gain: 5.500000
PMT 2 Gain: 5.000000
PMT 3 Gain: 5.500000

Meanwhile, the column headings are found in the second row which are explained in Section 10.2 Data Conversions (non-BRE) of ACME-DOC-010 Rev D, which begins on page 128. The basic conversions of the 125-Hz data to engineering units are provided in this section, but information on temperature conversions are provided in Sections 10.4 and 10.5 which are found in the same document on pages 142-143. Note that s-Flame did not feature the electric field used for the E-FIELD Flames investigation, so the data in columns IonCurrent, EfieldVoltage, and EfieldSetpointOut should be ignored.

As an example, the column headings from 20190A1_data.csv include the following (again displayed in two columns rather than a long row):

Time	TCBurner
MFCA1StaticMixerFlow	Radiometer+60
MFCA2BypassFlow	Radiometer+60CJ
MFCA3FuelFlow	Radiometer+30
IonCurrent	Radiometer+30CJ
PTA1	RadiometerCenter
PTA2	RadiometerCenterCJ
PTA3	Radiometer-30
EfieldVoltage	Radiometer-30CJ
PMT1	Radiometer-60
PMT2	Radiometer-60CJ
PMT3	RadiometerOffset
TCA	RadiometerOffsetCJ
TCB	TCE
TCC	Ch20
TCD	Ch21
TC200A	TCCJ
TC200B	EfieldSetpointOut

Note that the chamber pressure (PTA3 from this file) should be more accurately available from the CIR chamber calculator file discussed in section 4.C1. The ACME pressure transducers have not been calibrated in years, while the calibrations for the corresponding CIR pressure transducers are checked every 6 months.

Additional conversion is required for the measurements of the gas flow rates found in second through fourth columns of the data file. By column (e.g., in Excel), the flow rates are for:

B	MFCA1StaticMixerFlow	oxidizer (or inert) gas
C	MFCA2BypassFlow	nitrogen
D	MFCA3FuelFlow	fuel

The ACME MFCs are Hastings HFC-D-302 with default standard temperature and pressures (STP) of 0 Celsius and 760 torr for computations of volumetric flow or velocity. It is very important to understand that slpm, i.e., standard liters per minute, is a measurement of the mass (or alternatively molar) flow rate which is independent of the actual temperature or pressure. This is discussed for example in the MFC manual, where chapter 9 (on page 39) is "Volumetric vs. Mass Flow." Hastings's online converter at https://massflowconverter.com/ may be helpful as a check in this regard.

Conversions for the actual gases can be made with information provided in the manual on subsequent pages and the spreadsheets GasDataTables.xls and Inside gas mix calculator.xls which were both provided by the MFC manufacturer. First, open GasDataTables.xls and enable content, and then open Inside gas mix calculator.xls and enable content. From this, one should be able to enter mixtures and compute appropriate Gas Conversion Factors (GCFs).

Alternatively, the conversions above can be made with spreadsheets prepared by the ACME engineering team, where options include the following:

ACMEFlowCalculator.xlsx
ACMEFlowAndPressuresRawDataConversion.xlsx

2. Mass Flow Controller (MFC) Setpoint File

While the basic data file includes measurement of the gas flow rates, it may be helpful to understand the intended flow rates, where that information is found in a separate file for each test with a name like 20190A1_mfcSetpoints.csv. While it shares strong similarities to the basic data file, the rows of data are not added at a continuous and consistent rate. The file is briefly discussed in section 8.2 Mass Flow Controller Setpoint File on page 124 of ACME-DOC-010 Rev D.

3. Gas Chromatograph Instrumentation Package (GCIP) data

Measurements of the composition of the chamber's atmosphere were made using the GCIP on some s-Flame test days, typically either at the end of the day and/or the end of a set of tests. The raw data files are in a text (i.e.., ASCII) format and have names such as moduleDataACD_01_20200709_0125.asc for which a starting excerpt is shown here:

```
Version:            3
Maxchannels:    3
Sample ID:          1
Data File:          C:\EZIQ\Projects\Default\Data
Method:             C:\EZIQ\Projects\Default\Method\GCIP_Ascii_Method.met
User Name:      FCFOPS
Acquisition Date and Time:          _01_20200709_0125
Sample Rate:    100       100       100
Total Data Points:          25500   25500   25500
X Axis Title:       Seconds             Seconds             Seconds
Y Axis Title:       microvolts_(uV) microvolts_(uV) microvolts_(uV)
X Axis Multiplier:          0.01      0.01      0.01
Y Axis Multiplier:          10        100       100
-210
-198
-189
-183
-180
-176
-173
-167
```

-160
-155

These files can be analyzed with the INFICON's EZ IQ software for its 3000 Micro GC (version 3.3.2 SP2) which only works under Windows 7 or XP SP3 (with the NET Framework 3.0 and/or 3.5). INFICON has indicated that their newer software for the Micro GC Fusion will not work with 3000 Micro GC files. However, it is possible that software developed by Agilent may be able to work with these files, where they apparently developed the EZ IQ software initially.

NASA plans to analyze the files to determine species concentrations and make those results available, but the agency abandoned use of Windows 7 and earlier operating systems which has thus far prevented the planned analysis. As a result, species concentration results are not yet available from the GCIP measurements, where the instrument was not used in the experiments preceding ACME. It has furthermore only been used in a fraction of the ACME experiments, but s-Flame is one for which data was collected.

4. Space Acceleration Measurement System (SAMS) data

The Combustion Integrated Rack (CIR) in which the experiment was conducted, was equipped with a Passive Rack Isolation System (PaRIS), but it was not utilized for ACME testing because of the need for crew time to enable and disable the isolation (by respectively removing and installing alignment guides) on each test day. However, the CIR was equipped with a 3-axis SAMS sensor head. The acceleration archives, along with relevant documentation, are accessible from https://gipoc.grc.nasa.gov/wp/pims/home/, the home page for the Principal Investigator Microgravity Services (PIMS).

The following guidance for the SAMS data was provided by Kenneth Hrovat (NASA Glenn): In addition to the documentation and links from the PIMS home page, we introduce a few ways below to get started with exploring the CIR microgravity environment taking the perspective of SAMS measurements from within the rack.

Roadmap Spectrograms

To get a high-level, spectral roadmap view within two days after measurement, you can:

a. Use your browser and start at this link: http://pims.grc.nasa.gov/roadmap
b. In the upper left pane, you can navigate by GMT date to see a list of plots (PDF files) in the right pane for the GMT date of your choice. We refer to these PDFs as roadmap plots. For SAMS, these are acceleration spectrograms, which for vibratory data are a good way to look at patterns, structures and boundaries in both time and frequency for long time spans -- nominally, 8 hours per plot. These spectrograms are mostly qualitative and crudely quantitative three-dimensional plots in that a color scale attempts to map power spectral density magnitude as a color for the 3rd dimension. *NOTE*: Ticking the Navigation Filter for "es05" in the bottom pane will then only highlight days in yellow in the upper left pane when the operations team powered on the SAMS sensor in the CIR to take measurements.
c. You might want to navigate directly to a GMT day of interest for your science operations or run, but you can also browse day-by-day too in order to get some context comparing various 8-hour spans to get a feel for the environment under various conditions.

Binary Data

To download actual SAMS measurement data for use with your models or to correlate acceleration measurements with your science or support data, you can…

a. Use your browser and start at this link: http://pims.grc.nasa.gov/ftp/pad
b. Navigate by GMT date, for example you can follow year, month and day links to end up at: https://pims.grc.nasa.gov/ftp/pad/year2020/month06/day18/ where you see a list of SAMS sensor directories. The sensor subdirectory of interest for CIR is samses_accel_es05.
c. Once you click into that directory link, you will see data and header files listed for the CIR sensor.
d. The instructions at the following link help you to read these data: http://pims.grc.nasa.gov/plots/user/mohan/how2read_pad_binary_files.html

B. Image Files

Image data and files discussed in this section are associated with the following cameras:

Analog, operations (ops), internal camera
ACME Data Camera (DataCam)
Cool Flames Investigation (CFI) intensified camera
Low Light Level Ultraviolet (LLL-UV) camera
High Bit-depth Multi-spectral (HiBMs) camera and CIR illumination package

1. Analog, internal, operations (ops) camera

This color analog camera is mounted on the ACME chamber insert, and as such is within the combustion chamber and is sometimes referred to as the internal camera. Its primary purpose is to support the operations, e.g., to visualize the burner and igniter, and as such is also often referred to as the operations or ops camera.

The NTSC video (named for the National Television System Committee) from the ops camera is at 29.97 frames per second (fps) or nominally 30 fps. It is not recorded on the space station but is streamed to the ground in near real time and is recorded on CIR consoles at the Glenn ISS Payload Operations Center (GIPOC) in a Transport Stream (TS) format using VLC Media Player which can be freely downloaded from https://www.videolan.org/vlc/. That software has a snapshot function allowing for convenient extraction of still images, e.g., for making measurements of flame dimensions.

2. ACME Data Camera (DataCam)

ACME's primary camera is a high-definition (1360 x 1024), 12-bit color camera mounted outside the chamber on CIR's optical bench. It is equipped with a motorized lens with independent control of the zoom, iris, and focus. It has a filter barrel with a fused silica window (for color imaging), a blue-green BG7 Schott glass filter, and bandpass filters for imaging CH* and CH_2O (formaldehyde) chemiluminescence.

It was characterized prior to launch to enable temperature measurements via soot pyrometry and Thin Filament Pyrometry (TFP), soot volume fraction, and quantitative determination of the CH* concentration. However, the flames in s-Flame testing were soot-free with the

exception of ignition in some tests.

Soot temperatures can be determined from this camera's images via techniques such as those described in the following references:

a. S. De Iuliis, M. Barbini, S. Benecchi, F. Cignoli, and G. Zizak, Determination of the Soot Volume Fraction in an Ethylene Diffusion Flame by Multiwavelength Analysis of Soot Radiation, Combustion and Flame 115 (1998) 253–261. https://doi.org/10.1016/S0010-2180(97)00357-X

b. B. Ma and M. B. Long, Combined Soot Optical Characterization Using 2-D Multi-Angle Light Scattering and Spectrally Resolved Line-of-Sight Attenuation and Its Implication on Soot Color-Ratio Pyrometry, Applied Physics B 117 (2014) 287–303. https://doi.org/10.1007/s00340-014-5834-x

c. B. Ma, S. Cao, D. Giassi, D. P. Stocker, F. Takahashi, B. A. V. Bennett, M. D. Smooke, and M. B. Long, An Experimental and Computational Study of Soot Formation in a Coflow Jet Flame under Microgravity and Normal Gravity, Proceedings of the Combustion Institute 35 (2015) 839–846. https://doi.org/10.1016/j.proci.2014.05.064

d. N. J. Kempema, B. Ma, and M. B. Long, Investigation of In-Flame Soot Optical Properties in Laminar Coflow Diffusion Flames Using Thermophoretic Particle Sampling and Spectral Light Extinction, Applied Physics B 122 (2016). https://doi.org/10.1007/s00340-016-6509-6

e. N. J. Kempema and M. B. Long, Effect of Soot Self-Absorption on Color-Ratio Pyrometry in Laminar Coflow Diffusion Flames, Optics Letters 43 (2018) 1103. https://doi.org/10.1364/OL.43.001103

Meanwhile, TFP measurements can be made using approaches such as those discussed in:

f. V. Vilimpoc and L. P. Goss, SiC-Based Thin-Filament Pyrometry: Theory and Thermal Properties, Symposium (International) on Combustion 22 (1989) 1907–1914. https://doi.org/10.1016/S0082-0784(89)80205-X

g. W. M. Pitts, Thin-Filament Pyrometry in Flickering Laminar Diffusion Flames, Symposium (International) on Combustion 26 (1996) 1171–1179. https://doi.org/10.1016/S0082-0784(96)80333-X

h. J. D. Maun, P. B. Sunderland, and D. L. Urban, Thin-Filament Pyrometry with a Digital Still Camera, Applied Optics 46 (2007) 483–488. https://doi.org/10.1364/AO.46.000483

i. P. B. Kuhn, B. Ma, B. C. Connelly, M. D. Smooke, and M. B. Long, Soot and Thin-Filament Pyrometry Using a Color Digital Camera, Proceedings of the Combustion Institute 33 (2011) 743–750. https://doi.org/10.1016/j.proci.2010.05.006

j. B. Ma, G. Wang, G. Magnotti, R. S. Barlow, and M. B. Long, Intensity-Ratio and Color-Ratio Thin-Filament Pyrometry: Uncertainties and Accuracy, Combustion and Flame 161 (2014) 908–916. https://doi.org/10.1016/j.combustflame.2013.10.014

Particularly when making quantitative measurements, implementation of an Abel transform or similar is likely appropriate given the axisymmetry of the flames. In fact, all six of the

current ACME experiments feature axisymmetric or spherically symmetric flames. Approaches are compared in this paper:

k. C. J. Dasch, One-Dimensional Tomography: A Comparison of Abel, Onion-Peeling, and Filtered Backprojection Methods, *Applied Optics*, 31 (1992) 1146–1152. https://doi.org/10.1364/AO.31.001146

The a-k references above were taken from the following conference paper on early ACME results from the Coflow Laminar Diffusion Flame (CLD Flame) experiment:

Jesse Tinajero, Davide Giassi, Dennis Stocker, and Marshall Long. "Experimental Study on the Influence of Gravity on Highly Diluted and Sooting Coflow Flames," 11th US National Combustion Meeting of the Combustion Institute, Pasadena, CA, 24-27 March 2019.

Meanwhile, this paper on an ISS precursor to the CLD Flame experiment is an appropriate reference for determination of CH* concentration.

D. Giassi, S. Cao, B.A. Bennett, D.P. Stocker, F. Takahashi, M.D. Smooke, and M.B. Long, "Analysis of CH* Concentration and Flame Heat Release Rate in Laminar Coflow Diffusion Flames under Microgravity and Normal Gravity," *Combustion and Flame*, Vol. 167, pp. 198-206 (2016). https://doi.org/10.1016/j.combustflame.2016.02.012

Images are normally downlinked within a week of the testing, but selected images are often downlinked during a test day. Regardless, the images are downloaded in a wrapped Halcon object (HOBJ) format, which is not readily opened with typical image processing software. The image data was made available in that raw format and later in a converted TIFF format, where the NASA conversion was carried out without each of the following:

- color balancing (to avoid possible saturation as subsequently discussed)
- dark frame subtraction
- removal of hot spots from dead pixels (e.g., resulting from radiation damage)

Furthermore, the NASA conversion employs a standard bilinear demosaicing process which is detrimental when determining temperature profiles via Thin Filament Pyrometry (TFP) as it will average pixels both on and off the glowing fibers. The priority in the NASA conversion was to make the images more readily accessible, not to meet all needs.

Most ACME investigators are processing the raw HOBJ files using the OMA2 software developed by Prof. Marshall Long (Yale U.), the lead for ACME's CLD Flame experiment. The OMA2 software is freely available in both Mac and cross-platform (Qt) versions, where the latter can be used with both PC and Linux operating systems. A tutorial was prepared by Dr. Jesse Tinajero on the use of OMA2 to process ACME's HOBJ image files, where it can be found at https://www.youtube.com/watch?v=RMtu3KEXNiI. It has also been broken up and posted in three parts in which the image resolution may be better:

Part 1 https://www.youtube.com/watch?v=kdQHeQwokvY
Part 2 https://www.youtube.com/watch?v=Mua_8CfqhfM
Part 3 https://www.youtube.com/watch?v=BGRqgbIxZL4

The OMA2 read-hobj function was also rewritten by Tinajero as a Python function, where Python is an open source programming language that is easily obtainable for MacOS, Linux, and Windows operating systems. The Python script to open the HOBJ files can be downloaded from https://github.com/mblong/hobjExample. A tutorial for this function may be found at

https://www.youtube.com/watch?v=4wbgWeJSilc.

The HOBJ files can also be readily viewed and processed using ImageJ or its preloaded Fiji variant, where both can be freely downloaded. In either software package, the HOBJ files can imported (rather than opened) as raw files with the following settings:

Image type	16-bit unsigned
Width	1360
Height	1024
Offset to first image	6244

However, imported DataCam images will be monochrome unless a Bayer filter (e.g., as described at https://en.wikipedia.org/wiki/Bayer_filter) is subsequently implemented. It should be possible to install a plug-in to accomplish that color conversion. As another option, Python and Matlab both have image demosaicing functions.

To use the ACME DataCam for quantitative measurements, it is critical to avoid saturation, for example by using a camera gain of zero. Another employed approach was to not use the camera's white balance which by default multiplies the red and blue by 1.41 and 1.68, respectively, because that could inadvertently lead to saturation. Such multiplication can be done after the HOBJ file is converted and the colors demosaiced, where in OMA2 the white balancing is accomplished with the commands:

```
*rgb 1.41 1 1.68
clip 4095
```

In summary, the white balance, if desired, needs to be done after the fact through software because it was at least generally not done with the hardware to avoid the risk of saturation.

However, in carrying out color-ratio pyrometry calculations for the ISS data from ACME's CLD Flame experiment, the investigators determined that the three color ratios (e.g., blue/red) didn't result in the same temperature profile until the green channel was adjusted by 10%. This notably differs from the camera's default of multiplication by 1 (i.e., no multiplication) for the green color plane.

For Thin Filament Pyrometry (TFP): "After rigorous testing, it was determined [by Tinajero and Long] that a modified version of the bilinear method could be used to optimize the conversion of raw fiber images to full resolution color images. This modified version biases the linear interpolation along the length of the fiber. This method combined with moderate use of defocusing was found to reduce artifacts created by standard demosaicing algorithms and was used in all of the fiber measurements described below."

> Jesse Tinajero and Marshall Long. "Dilution Limits of Coflow Laminar Methane and Ethylene Flames in Microgravity Versus Normal Gravity," 12th U.S. National Combustion Meeting, 24-26 May 2021.

Note that the ACME images will be naturally dark because of the use of the 16-bit file format with 12-bit image data. As such, adjustments to the image brightness will generally be necessary even if the images are not demosaiced. Furthermore, cameras' CCD arrays are susceptible to radiation damage while in orbit. As a result, the images feature hot pixels which increase in number over the months and years in space. As such, it will appear as if there is a star field within the combustion chamber. Therefore, the images should ideally be processed

to remove the bright outlying hot spots. Radiation damage is also obvious in the ops camera imaging.

Although the DataCam was typically set for imaging at 30 fps, that is trumped by the exposure settings which were often long because of the dim nature of the microgravity flames. As a result, the framing rate was typically far less. Different exposures are often selected for different phases of the tests, e.g., prior to ignition and during the test, so the framing rate may not be consistent. Bracketed imaging could also be employed, where a repeating sequence of different exposure durations is used to enable appropriate photography under uncertain or varying flame intensities. Bracketing can also used to simultaneously accomplish multiple imaging objectives, e.g., to appropriately image both brightly glowing soot (or TFP fibers) as well as the much dimmer blue portion of the flame.

In interpreting the image data, it may be helpful to check the ACME DataCam metadata file which should be available for each successful flame test, e.g., to verify the exposure duration. These files are in a comma-delimited (CSV) format and can be opened with spreadsheet software or alternately as a text file. Column headings are found in the first row of such files.

ACME DataCam metadata

This is an example of the metadata format from ACME_20190A1.mdipsug. The column headings are presented with the values for the first image (i.e., from the file's second row).

Timestamp	20200708_135433.120
Filename	/sd1/sci/20190A1/20200708_135433.120_ACME_20190A1_00000.hobj
DeltaMilliseconds	236
CalculatedRate	0
BinningX	1
BinningY	1
BlockId	28
BufferIncomplete	0
BufferIncompleteTotal	0
BuffersNum	5
ExposureManual	100000
FrameRate	30.0003
GainManual	0
Height	1024
PixelFormat	BayerRG12
RegionX	0
RegionY	0
TickTimestamp	1.11931E+13
Width	1360

Again, it is very important to understand that the exposure time, in microseconds, trumps the specified framing rate (in fps), i.e., where long exposures will lead to a shorter actual framing rate. In this example case, the 0.1-s exposure time means that the framing rate (at the test's start) was nominally 10 fps rather than the 30 indicated by FrameRate. It is furthermore important to understand that the exposure time (and thus actual framing rate) was often varied during the test, where that could be seldom or often if bracketing was employed, i.e.,

where a repeated sequence of images was taken at different exposures.

Unfortunately, it has been seen during bracketing that the reporting of exposure times in the metadata files is, or at least can be, offset. Therefore, it is important to identify the brightest images, for example, in an image sequence to determine which had the longest of the bracketed exposure times.

The parameter files, with names like 'as run20190A1.txt,' control the DataCam exposures and much more. However, these files have not always saved as intended and were sometimes corrected after the fact. In contrast, the text-file logs, with names such as 2020-06-24_15-10_ACME.log, are a reliable source for the parameters which specified the imaging settings and much more. There is a single file of this type for each test day (at least normally) and the parameters are a small fraction of the automated log content, but one can search for a test by its number. Log files like this are discussed in section 4.C.5 of this document.

Due to the use of a turning mirror, the DataCam images should be flipped vertically if upright images are desired, i.e., where the burner is at the bottom and the flame extends upward. In this regard, note that the DataCam provides images in a 'landscape' format with the image's long dimension orthogonal to the burner and chamber axis. This is contrary to an ACME science requirement specifying that the long dimension be aligned with the experimental axis to provide images in a 'portrait' format. In this regard, the science requirements were not always met, e.g., because of design tradeoffs.

There is some variation in the ACME camera zoom, where its setting is not completely reproducible. Therefore, the scaling of the image data should ideally be established for each subset of tests if more than one zoom setting was used within a test day. The scaling can typically be carried out using images regularly taken of the ceramic fibers for Thin Filament Pyrometry (TFP), where such imaging is normally made at two fiber array positions. Meanwhile, the other cameras used in ACME testing had fixed lenses.

3. Cool Flames Investigation (CFI) intensified camera

This intensified camera was typically configured with a CH* bandpass filter and was generally archived at 23 fps. However, s-Flame tests were sometimes instead conducted with a formaldehyde (CH_2O) filter or a clear fused-silica filter in the case of the hydrogen flames in the second-round testing (i.e., with helium dilution). The filter is normally identified in files with names such as 20190 test sheets.pdf.

The grayscale image files are in a Tagged Image File Format (TIFF) format and can be opened readily, where brightness/contrast adjustment may again be required.

Note that this camera's Field of View (FOV) is partially blocked by the arms of the fiber array for Thin Filament Pyrometry (TFP) when it is moved to position the fibers across the flame.

As with the ACME DataCam, it might be useful to check the metadata for the CFI intensified camera, which is in a comma-delimited (CSV) format and can be opened with spreadsheet software or alternately as a text file.

CFI intensified camera metadata

An example file name is INTNS_20190A1.mdipsug where INTNS designates the CFI intensified camera. Presented below are the column headings and example values from the test's first image (numbered 000000), and the corresponding camera parameter name and a description:

Timestamp 20200708_135433.016
Camera parameter: n/a
The timestamp portion of the filename, when the image was grabbed from the camera.

Filename /sd1/sci/20190A1/20200708_135433.016_INTNS_20190A1_00000.tiff
Camera parameter:
Image path and file name with timestamp of when the image was grabbed from the camera

DeltaMilliseconds 132
Camera parameter: n/a
Difference between last image timestamp and current image timestamp.

CalculatedRate 0
Camera parameter: n/a
This is GevTimestampTickFrequency / (GtlBufferTimestamp - GtlBufferTimestamp of last image). GevTimestampTickFrequency is querried each image.

BinningX x1
Camera parameter: BinningHorizontal
The number of horizontal photo-sensitive cells that must be combined (added) together. Note that the user has to set the Width feature accordingly.

BinningY x1
Camera parameter: BinningVertical
The number of vertical photo-sensitive cells that must be combined (added) together. Note that the user has to set the Height feature accordingly.

BitShift NoShift
Camera parameter: BitShift
The number of bits to shift in the data output.

BlockId 2
Camera parameter: GtlBlockID
ID of the last received block (typically of type image) as reported by the device. If no block has been received, 0 is returned.

BufferIncomplete 0
Camera parameter: GtlIncompleteBuffers
Current number of incomplete buffers received (where the Camera Parameter should be GtlBufferIncomplete but is GtlIncompleteBuffers instead).

BufferIncompleteTotal 0
Camera parameter: GtlIncompleteBuffers
Current number of incomplete buffers received.

BuffersNum 5
Camera parameter: GtlNumBuffers
Number of currently used buffers for the internal ring buffer.

CameraSpeed Normal
Camera parameter: CameraSpeed

The camera speed.

ExposureAuto 43067

Camera parameter: CurrentAecExposure

The current AEC exposure value.

ExposureManual 32000

Camera parameter: ExposureTimeRaw

The Exposure time in units of uSeconds when ExposureMode is Timed (and AEC is disabled).

ExposureMode 0

Camera parameter: AecEnable

Whether or not the Automatic Exposure Control function is enabled.

FrameRate 23

Camera parameter: CurrentFrameRate

The current frame rate in integer units of frames per second. This is a dynamic value that changes as a function of the camera's mode of operation.

GainAuto 143

Camera parameter: CurrentAgcGain

The current AGC analog gain value.

GainManual 143

Camera parameter: GainRaw

The selected analog gain as a raw integer value (when AGC is disabled).

GainMode 0

Camera parameter: AgcEnable

Whether or not the Automatic Gain Control function is enabled.

ImageHeight 1040

Camera parameter: SensorHeight

The effective height of the sensor in pixels.

ImageWidth 1392

Camera parameter: SensorWidth

The effective width of the sensor in pixels.

PixelFormat Mono16

Camera parameter: PixelFormat

The format of the pixel to use during the acquisition.

RegionHeight 1040

Camera parameter: Height

The actual image height expelled by the camera of the master AOI (in pixels).

RegionWidth 1392

Camera parameter: Width

The actual image width expelled by the camera of the master AOI (in pixels).

RegionX 0

Camera parameter: OffsetX

The horizontal offset from the origin to the master AOI (in pixels).

RegionY 0

Camera parameter: OffsetY

The vertical offset from the origin to the master AOI (in pixels).

TickTimestamp 21607337875

Camera parameter: GtlBufferTimestamp

Time stamp of the current buffer in ticks. The time in seconds can be calculated via the parameter 'GevTimestampTickFrequency'.

4. **Low Light Level Ultraviolet (LLL-UV) camera**
   This monochrome camera has an intensifier (for 280-700 nm) and a 310-nm bandpass filter (with a 10-nm FWHM) for OH* imaging. As a result of the signal loss through the filter, binning is employed such that the OH* images are recorded at 512x512 and 30 fps. After post-processing on the ground, the grayscale image files are available in a Tagged Image File Format (TIFF) format and can be opened readily, where brightness/contrast adjustment may be required.

   Like the CFI intensified camera, the LLL-UV camera's Field of View (FOV) is partially blocked by the arms of the fiber array for Thin Filament Pyrometry (TFP) when it is moved to position the fibers across the flame.

   While the LLL-UV camera was used throughout s-Flame testing, it wasn't used at all for the other five ACME experiments.

5. **High Bit-depth Multi-spectral (HiBMs) camera and CIR illumination package**
   The HiBMs camera was used in combination with an opposed backlight, i.e., the CIR's collimated illumination package. The resultant imaging was used in preceding droplet combustion experiments for measuring droplet diameter, but it may also be useful for determining the soot volume fraction via light extinction. However, the camera's red filter, corresponding to the red illumination, appears to be broad enough to permit significant soot incandescence to pass through the filter. As a result, this imaging is unlikely to be useful for soot volume fraction measurements for all conditions. After post-processing on the ground, the images were made available in the TIFF format and as such are readily opened. The grayscale images were typically archived at 15 fps.

   It is important to understand that this camera's images are of little value for s-Flame because of the general lack of soot. However, there was sometimes soot seen during ignition. neglecting ignition in some cases. But even considering that, the soot volume fraction can be computed using images from the ACME DataCam as discussed in that camera's section.

   In initial ACME testing with the Coflow Laminar Diffusion Flame (CLD Flame) experiment, this monochrome camera was used without a filter, but that was abandoned because the camera is nearly insensitive to blue. The red filter was installed on 5 March 2018 and used throughout all subsequent ACME testing.

C. Test Parameters and Other Files

Files discussed in this section include the following:

- CIR chamber calculator file
- Test parameter file
- Notes
- Test timing
- Logs

1. CIR chamber calculator file

In great contrast to other file types, there is a single file, named 'ACME Chamber Calculator - 2018-03-08 to 2020-07-08.xlsx,' which provides estimates of the composition of the combustion chamber before and after each s-Flame test (and during earlier testing), and more generally for every Fuel/Oxidizer Management Assembly (FOMA) action involving gas flow into or out of the chamber. As a result, this includes filling and venting operations, plus recirculation of the chamber contents through the adsorber cartridge filled with 13x molecular sieves.

More specifically, the chamber calculator provides the chamber pressure before and after each operation involving flow into or out of the chamber. While the standard ACME data output files (discussed in section 4.A.1) provide the chamber pressure throughout each test, that measurement is from a pressure transducer that hasn't been calibrated in years. Meanwhile, the calibrations of the corresponding CIR pressure transducers are checked every six months and if needed those calibrations are updated. Therefore, the chamber calculator is the more reliable source of the chamber pressure.

The pressure in the CIR chamber is measured by four different CIR pressure transducers, of which one (PT29) is a different model used for safety. The output of the other three transducers were used to obtain an average chamber pressure. However, the output of one of those (PT28) was clearly an outlier from the other two (PT16 and PT17) as well that of PT29. Therefore, the average pressure recorded in the chamber calculator file was later based on just PT16 and PT17.

For s-Flame, the chamber calculator also provides concentration estimates of the following gases within the chamber before and after each test and FOMA operation:

Oxygen & inerts:	O_2, N_2, He
Fuels:	H_2, CH_2, C_2H_4
Combustion products:	CO_2, H_2O

The latter are based on telemetry from the three Mass Flow Controllers (MFCs) on the ACME chamber insert for the flow of fuel, oxygen (or inert), and nitrogen.

In a test, the chamber calculator assumes either complete combustion or no combustion, where in the former case the reactant flowing from the burner is transformed into carbon dioxide and water vapor (or water vapor alone if the fuel was hydrogen). While failed or unsustained ignitions were typically set to assume no ignition. Short-lived flames were often treated as having complete combustion, even if a significant fraction of the reactant didn't burn. As such, the chamber calculator has its limitations.

With recirculation of the chamber contents, the calculator assumed that the carbon dioxide and water vapor were completely adsorbed, i.e., removed from the chamber. Meanwhile, it assumes that any unburnt fuels – hydrogen, methane. and ethylene - are not adsorbed.

In Aug. 2021, it was determined that the ACME chamber calculator doesn't account for gas conversion factors where the Mass Flow Controllers (MFCs) were calibrated for nitrogen. This shortcoming can affect the results significantly. As an example, the conversion factor for ethylene - the most commonly used ACME fuel - is nominally 0.6 which is quite different from

1. See section 4.A.1 of this document for additional information about this conversion.

At the time of this writing, NASA is working on improvements to the chamber calculator file, but researchers can of course make their estimates of the chamber contents where the chamber calculator should nonetheless be referenced for its record of fills, venting, and recirculation. Note that the chamber's free volume is approximately 83 liters depending on the installed hardware. Unfortunately, it is not yet possible to compare chamber calculator estimates with measurements that were sometimes made with the CIR's Gas Chromatograph Instrumentation Package (GCIP). Concentrations from the latter are not yet available as explained in section 4.A.3 of this document.

2. Test parameter file

Each s-Flame test was controlled by a software script and corresponding parameter file, named for example 'as run20176B1.txt,' specifying settings for imaging, ignition, gas flow, and more. The parameter file and script are discussed extensively in the ACME/s-Flame Experimental Procedure document, which should be referred to for information on this file type. But note that the ACME.log files should be the best way to verify the as-run test parameters, where the separate parameter files are unfortunately not completely reliable. They have not always been saved and when the file saving was automated, they have still sometimes had missing entries.

3. Notes

Notes of various types are available for test days, where not all types were collected or necessarily prepared for each test day. It is important to understand that the notes, whether handwritten or typed, may include mistakes. It is better to rely on automated logs, metadata files, and parameter files or of course the measurements themselves when possible.

The daily test point logs, with names such as '19254 daily test point log.pdf,' chronologically list the tests and the image data acquired which was to be downlinked. Typically, image data was not downlinked without successful, i.e., sustained, ignition. As an example, image data wasn't downlinked if the test aborted prior to ignition. However, the reverse is not always true. Image data was sometimes downlinked when ignition was unsuccessful and as a result the test point log is only a clue as to what transpired. Furthermore, note that image data from certain selected cameras, e.g., the HiBMs camera, was sometimes not downlinked due to the limit of how much data could be reasonably downlinked in a subsequent two-shift day.

Notes with names such as '19254 flow traces etc.pdf' include preliminary plots of the flow rates (as seen on a console during testing) and other flame diagnostics, typically along with additional information about the tests.

The test parameter notes, with names such as '19254 test parameter notes.pdf,' show the parameter values which were planned in advance of the test day and the changes made during the test day.

The test sheets, with names such as '19254 test sheets.pdf,' are summary notes about the conduct of each tests conducted that day.

4. Test timing

Timing files with names such as 20190A1_TP.txt can be a convenient way to determine the phases of the test in which the images were acquired. An example s-Flame file is presented here. Regarding the closing table, it should be understand that IPSU1 and IPSU2 were respectively used to archive the HiBMs and LLL-UV images, while IPSU5 (also known as IPSU-G) was used to archive the ACME DataCam and CFI intensified camera images.

Test Point: 20190A1

Script start: 2020-07-08 13:53:32
Script paused #1: 2020-07-08 13:53:42
Script resumed #1: 2020-07-08 13:54:18
Pre-ignition data: 2020-07-08 13:54:18
Pre-ignition images: ***
Preheat igniter: 2020-07-08 13:54:49
MFCs on for priming: ***
Ignition and retraction: 2020-07-08 13:54:51
Flame stabilization #1: 2020-07-08 13:54:51
Dilution of fuel and inert: ***
Flame stabilization #2: 2020-07-08 13:54:51
Ramp coflow to match test cond: ***
Flame stabilization #3: ***
Ramp to final velocity: ***
Constant flow: ***
MFCs off: 2020-07-08 13:54:53
Post-flame images: ***
Post-flame data: 190 13:54 66~zt.sc
Script complete after abort: 2020-07-08 13:55:16
Extinction detected: 2020-07-08 13:54:53
Script paused #2: ***
Script resumed #2: ***
Test Point Failure: 2020-07-08 13:54:53
Reason: EXTINCTION DETECTED

Camera	Archived	Start Image	End Image
IPSU1	623	0	623
IPSU2	263	0	263
ACME	176	10	176
Intensified	577	0	577

5. Logs

There are four primary types of automated logs of which two, with names like the following, may be helpful.

2020-06-24_15-10_ACME.log
2020-06-24_15-00_IPSUG.log

As a reminder, the IPSU-G is the processing and storage unit for the images from the ACME

DataCam and CFI intensified camera.

Meanwhile, the two logs of commands, with names like the following, are unlikely to be useful for interpreting the s-Flame results.

2020-06-24_15-30_ACMECommands.log
2020-06-24_15-05_IPSUGCommands.log

Each of the four log files are created for each day of testing, rather than for each test, although there are sometimes multiple log files (of the same type) covering separate portions of a test day. They are text files (despite the log suffix) and as such can be readily opened.

The ACME.log files should be the best way to verify the test parameters, where the separate as-run parameter files are unfortunately not completely reliable. A portion of the log corresponding to a single test, 20176B1, is presented here, where this can best be interpreted after becoming familiar with the contents of the ACME/s-Flame Experimental Procedure document. But the test parameters and timing of the test are obvious in the excerpt.

ACME.log excerpt

```
2020-06-24 176 17:40:40:687~acme~"Received command: SystemRunScript()"
2020-06-24 176 17:40:40:687~zt.script~param: "testpointName":"20176B1"
2020-06-24 176 17:40:40:687~zt.script~param: "preIgnitionDataTimeSec":0
2020-06-24 176 17:40:40:687~zt.script~param: "postFlameDataTimeSec":15
2020-06-24 176 17:40:40:688~zt.script~param: "useIPSUs":true
2020-06-24 176 17:40:40:688~zt.script~param: "ipsuTimeoutSec":40
2020-06-24 176 17:40:40:688~zt.script~param: "useAcmeCamera":true
2020-06-24 176 17:40:40:688~zt.script~param: "useIntensifiedCamera":true
2020-06-24 176 17:40:40:688~zt.script~param: "ipsu1PreIgnitionArchiveRate":100
2020-06-24 176 17:40:40:688~zt.script~param: "ipsu2PreIgnitionArchiveRate":100
2020-06-24 176 17:40:40:688~zt.script~param: "ipsu1TestArchiveRate":2
2020-06-24 176 17:40:40:688~zt.script~param: "ipsu2TestArchiveRate":1
2020-06-24 176 17:40:40:688~zt.script~param:
    "dataCameraPreIgnitionExposureUs":100000
2020-06-24 176 17:40:40:688~zt.script~param: "dataCameraIgnitionExposureUs":100000
2020-06-24 176 17:40:40:688~zt.script~param: "dataCameraTestExposureUs":650000
2020-06-24 176 17:40:40:688~zt.script~param: "dataCameraExtinctionExposureUs":650000
2020-06-24 176 17:40:40:688~zt.script~param: "analogCameraPreIgnitionShutter":2
2020-06-24 176 17:40:40:688~zt.script~param: "analogCameraIgnitionShutter":2
2020-06-24 176 17:40:40:688~zt.script~param: "analogCameraTestShutter":2
2020-06-24 176 17:40:40:688~zt.script~param: "analogCameraExtinctionShutter":0
2020-06-24 176 17:40:40:688~zt.script~param: "intensifierGainAmplitude":50000
2020-06-24 176 17:40:40:689~zt.script~param: "preIgnitionImageTimeSec":0
2020-06-24 176 17:40:40:689~zt.script~param: "postFlameImagingTimeSec":3
2020-06-24 176 17:40:40:689~zt.script~param: "useBracketedExposures":true
2020-06-24 176 17:40:40:689~zt.script~param: "bracketingImagesPerExposure":1
2020-06-24 176 17:40:40:689~zt.script~param: "bracketingNumberOfExposures":3
2020-06-24 176 17:40:40:689~zt.script~param: "bracketingExposureStartUsec":60000
2020-06-24 176 17:40:40:689~zt.script~param:
    "bracketingExposureMultiplierTimes100":500
```

2020-06-24 176 17:40:40:689~zt.script~param: "igniterIgnitionPosition":2250
2020-06-24 176 17:40:40:689~zt.script~param: "igniterPreheatPowerV":4.49
2020-06-24 176 17:40:40:689~zt.script~param: "igniterPreheatTimeMs":2000
2020-06-24 176 17:40:40:689~zt.script~param: "igniterIgnitionPowerV":4.49
2020-06-24 176 17:40:40:689~zt.script~param: "igniterIgnitionDurationMs":500
2020-06-24 176 17:40:40:689~zt.script~param: "igniterDelayBeforeRetractMs":500
2020-06-24 176 17:40:40:689~zt.script~param: "igniterRetractVelocityCountsPerSec":500
2020-06-24 176 17:40:40:689~zt.script~param: "useIgnitionDetection":true
2020-06-24 176 17:40:40:689~zt.script~param: "extinctionDetectionStopsTest":true
2020-06-24 176 17:40:40:689~zt.script~param: "pmt1Gain":4.5
2020-06-24 176 17:40:40:690~zt.script~param: "pmt2Gain":5
2020-06-24 176 17:40:40:690~zt.script~param: "pmt3Gain":5
2020-06-24 176 17:40:40:690~zt.script~param: "fiberIgnitionPosition":10000
2020-06-24 176 17:40:40:690~zt.script~param: "fiberTestPosition":14100
2020-06-24 176 17:40:40:690~zt.script~param: "fiberMoveDuringTest":false
2020-06-24 176 17:40:40:690~zt.script~param: "fiberSegmentDistance":400
2020-06-24 176 17:40:40:690~zt.script~param: "fiberVelocity":500
2020-06-24 176 17:40:40:690~zt.script~param: "fiberAcceleration":1000
2020-06-24 176 17:40:40:690~zt.script~param: "fiberDeceleration":0
2020-06-24 176 17:40:40:690~zt.script~param: "fiberHoldDurationSec":1
2020-06-24 176 17:40:40:690~zt.script~param: "fiberCycles":100
2020-06-24 176 17:40:40:690~zt.script~param: "useFoma":true
2020-06-24 176 17:40:40:690~zt.script~param: "flowTimeoutSec":120
2020-06-24 176 17:40:40:690~zt.script~param: "nitrogenPriming1DurationSec":0
2020-06-24 176 17:40:40:690~zt.script~param: "nitrogenPriming2DurationSec":0
2020-06-24 176 17:40:40:690~zt.script~param: "fuelFlowIgnitionDeltaMs":250
2020-06-24 176 17:40:40:690~zt.script~param: "stabilizationTime1Sec":0
2020-06-24 176 17:40:40:691~zt.script~param: "stabilizationTime2Sec":0
2020-06-24 176 17:40:40:691~zt.script~param:
"fuelBottleIgnitionFlowRateN2Slpm":0.60319
2020-06-24 176 17:40:40:691~zt.script~param: "nitrogenIgnitionFlowRateN2Slpm":0
2020-06-24 176 17:40:40:691~zt.script~param: "fuelBottleTestFlowRateN2Slpm":0.60319
2020-06-24 176 17:40:40:691~zt.script~param: "nitrogenTestFlowRateN2Slpm":0
2020-06-24 176 17:40:40:691~zt.script~param: "ignitionToTestRampTimeSec":0
2020-06-24 176 17:40:40:691~zt.script~param: "constantFlowTimeSec":45
2020-06-24 176 17:40:40:691~zt.script~param: "numberOfRampHolds":0
2020-06-24 176 17:40:40:691~zt.script~param: "rampSizeFuelSlpm":0
2020-06-24 176 17:40:40:691~zt.script~param: "rampSizeN2Slpm":0
2020-06-24 176 17:40:40:691~zt.script~param: "rampDurationSec":0
2020-06-24 176 17:40:40:691~zt.script~param: "holdDurationSec":0
2020-06-24 176 17:40:40:691~zt.script~param: "burnerTemperatureLimitC":250
2020-06-24 176 17:40:40:691~zt.script~param: "burnerTemperatureLimitDurationSecs":3
2020-06-24 176 17:40:40:691~zt.script~s-Flame test point 20176B1 STARTED
2020-06-24 176 17:40:40:691~zt.script~Setting PMT gains
2020-06-24 176 17:40:40:692~zt.script~Ignition Timing Values:
2020-06-24 176 17:40:40:692~zt.script~----------
2020-06-24 176 17:40:40:692~zt.script~Start of Fuel flow at ignition rate 1750 Ms

2020-06-24 176 17:40:40:692~acme.diag~"Received command:
PmtSetAllGains(PMT1Gain:4.5,PMT2Gain:5,PMT3Gain:5)"
2020-06-24 176 17:40:40:692~zt.script~Start of igniter preheat: 0 Ms
2020-06-24 176 17:40:40:692~zt.script~Start of ignition: 2000 Ms
2020-06-24 176 17:40:40:692~zt.script~Start of igniter retraction: 2500 Ms
2020-06-24 176 17:40:40:692~zt.script~Start of stabilization time 1: 2500 Ms
2020-06-24 176 17:40:40:692~zt.script~----------
2020-06-24 176 17:40:40:692~zt.script~temperature limit of 250 C is
9.226326175697157mV
2020-06-24 176 17:40:40:692~zt.script~Changing cameras to pre-ignition settings
2020-06-24 176 17:40:40:692~acme~"Received command:
SystemOpenTestPoint(testpointName:20176B1)"
2020-06-24 176 17:40:40:692~zt.script~Intensified camera to safe mode
2020-06-24 176 17:40:40:694~acme.image~"Received command:
DataCameraSetManualExposure(integrationTimeUs:100000)"
2020-06-24 176 17:40:40:695~acme.image~"Received command:
AnalogCameraSetShutter(shutter:2)"
2020-06-24 176 17:40:40:730~acme.image~Analog command successful.
2020-06-24 176 17:40:40:730~acme.image~"Received command:
IntensifiedCameraGate(gateType:0,pulsedGateWidthCounts:0)"
2020-06-24 176 17:40:40:916~zt.fcf~ -$$$$$$$ PRIORITY INVERSION:8e9d000 sendRvc 59
100
2020-06-24 176 17:40:41:193~acme.image~"Received command:
IntensifiedCameraGainAmplitude(gainAmplitudeCounts:2048)"
2020-06-24 176 17:40:43:193~zt.script~Moving fiber to its starting position
2020-06-24 176 17:40:43:193~acme.image~"Received command:
IlluminationConfigure(sideLEDBankState:1,endLEDBankState:1)"
2020-06-24 176 17:40:43:193~zt.script~Inserting igniter
2020-06-24 176 17:40:43:194~zt.script~Clearing MFC integrated flow
2020-06-24 176 17:40:43:194~acme.fiber~"Received command:
FiberMove(position:10000,velocity:3000)"
2020-06-24 176 17:40:43:453~acme.igniter~"Received command:
IgniterMove(position:2250)"
2020-06-24 176 17:40:43:712~acme.diag~"Received command:
ExtinctionDetectionSetState(state:0)"
2020-06-24 176 17:40:43:713~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*00g31=0)"
2020-06-24 176 17:40:43:914~acme.flow~Reply for command: *00g31=0
2020-06-24 176 17:40:43:914~acme.flow~">"
2020-06-24 176 17:40:44:194~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*01g31=0)"
2020-06-24 176 17:40:44:762~acme.flow~Reply for command: *01g31=0
2020-06-24 176 17:40:44:762~acme.flow~">"
2020-06-24 176 17:40:45:194~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*02g31=0)"
2020-06-24 176 17:40:45:719~acme.flow~Reply for command: *02g31=0
2020-06-24 176 17:40:45:719~acme.flow~">"

2020-06-24 176 17:40:46:195~zt.script~Reporting initial MFC integrated flow
2020-06-24 176 17:40:46:195~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*00g31)"
2020-06-24 176 17:40:46:722~acme.flow~Reply for command: *00g31
2020-06-24 176 17:40:46:722~acme.flow~"0
2020-06-24 176 17:40:46:722~acme.flow~>"
2020-06-24 176 17:40:47:195~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*01g31)"
2020-06-24 176 17:40:47:723~acme.flow~Reply for command: *01g31
2020-06-24 176 17:40:47:723~acme.flow~"0
2020-06-24 176 17:40:47:723~acme.flow~>"
2020-06-24 176 17:40:48:196~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*02g31)"
2020-06-24 176 17:40:48:720~acme.flow~Reply for command: *02g31
2020-06-24 176 17:40:48:720~acme.flow~"0
2020-06-24 176 17:40:48:720~acme.flow~>"
2020-06-24 176 17:40:49:196~zt.script~Test point PAUSED to check igniter.
SystemResumeScript when ready.
2020-06-24 176 17:40:50:196~acme~"Received command: SystemPauseScript()"
2020-06-24 176 17:40:50:196~zt.script~Script paused.
2020-06-24 176 17:41:10:950~acme~"Received command: SystemResumeScript()"
2020-06-24 176 17:41:10:950~zt.script~Script resumed.
2020-06-24 176 17:41:11:016~zt.script~Recording pre-ignition data for 0 sec
2020-06-24 176 17:41:11:016~acme.igniter~"Received command:
IgniterSetupMoveWithParameters(position:0,velocity:500,acceleration:5000,deceleratio
n:5000)"
2020-06-24 176 17:41:11:016~zt.script~Setting IPSU(s) pre-ignition archive rates
2020-06-24 176 17:41:11:116~zt.script~Starting IPSU(s) archiving. Waiting up to 40 sec
2020-06-24 176 17:41:11:161~acme.flow~"Received command: FlowCloseAll()"
2020-06-24 176 17:41:11:162~acme.diag~"Received command: DataRecordingStart()"
2020-06-24 176 17:41:11:162~acme.image~"Received command:
IpsuSetArchiveRate(ipsu:2,archiveRate:100)"
2020-06-24 176 17:41:11:163~acme.image~"Received command:
IpsuSetArchiveRate(ipsu:2,archiveRate:100)"
2020-06-24 176 17:41:11:163~acme.image~"Received command:
StartIpsus(testpointName:20176B1,ipsu5NumberCameras:2,ipsu5Camera1:3,ipsu5Came
ra2:4,ipsu5Camera3:0,ipsu5Camera4:0,ipsu6NumberCameras:0,ipsu6Camera1:0,ipsu6C
amera2:0,ipsu6Camera3:0,ipsu6Camera4:0)"
2020-06-24 176 17:41:19:731~acme.ipsu~Success for IPSU 1 command IpsuTestpointOpen
2020-06-24 176 17:41:25:473~acme.ipsu~Success for IPSU 2 command IpsuTestpointOpen
2020-06-24 176 17:41:25:512~acme.ipsu~Success for IPSU 5 command IpsugTestpointOpen
2020-06-24 176 17:41:25:537~zt.fcf~ -$$$$$$$ PRIORITY INVERSION:8e9d000 sendRvc 59
100
2020-06-24 176 17:41:25:538~acme.ipsu~Success for IPSU 1 command IpsuEnableArchive
2020-06-24 176 17:41:25:560~acme.ipsu~Success for IPSU 2 command IpsuEnableArchive
2020-06-24 176 17:41:25:568~zt.fcf~ -$$$$$$$ PRIORITY INVERSION:8e9d000 sendRvc 59
100

2020-06-24 176 17:41:25:569~acme.ipsu~Success for IPSU 1 command IpsuDisableLiveVideo
2020-06-24 176 17:41:25:879~acme.ipsu~Success for IPSU 2 command IpsuDisableLiveVideo
2020-06-24 176 17:41:25:899~zt.fcf~ -$$$$$$$ PRIORITY INVERSION:8e9d000 sendRvc 59 100
2020-06-24 176 17:41:25:900~acme.ipsu~Success for IPSU 1 command IpsuTestpointStart
2020-06-24 176 17:41:25:927~acme.ipsu~Success for IPSU 2 command IpsuTestpointStart
2020-06-24 176 17:41:25:930~zt.fcf~ -$$$$$$$ PRIORITY INVERSION:8e9d000 sendRvc 59 100
2020-06-24 176 17:41:25:931~acme.ipsu~Success for IPSU 5 command IpsugCamerasStartTestPointArchive
2020-06-24 176 17:41:27:120~zt.script~Recording pre-ignition images for 0 sec
2020-06-24 176 17:41:27:120~zt.script~Opening CIR fuel manifold
2020-06-24 176 17:41:27:121~acme~Sent IopRunScript command to IOP.
2020-06-24 176 17:41:42:227~acme~"Received command: SystemCirFlowState(fuelManifoldState:1,diluentManifoldState:0,oxygenManifoldState:0,nitrogenManifoldState:0)"
2020-06-24 176 17:41:42:253~zt.script~CIR Manifold State. Fuel:Open Diluent:Closed Oxygen:Closed Nitrogen:Closed
2020-06-24 176 17:41:42:253~zt.script~Intensified camera to continuous mode
2020-06-24 176 17:41:42:253~acme.image~"Received command: IlluminationConfigure(sideLEDBankState:0,endLEDBankState:0)"
2020-06-24 176 17:41:42:254~acme.image~"Received command: IntensifiedCameraGate(gateType:1,pulsedGateWidthCounts:0)"
2020-06-24 176 17:41:42:276~zt.fcf~ -$$$$$$$ PRIORITY INVERSION:8e9d000 sendRvc 59 100
2020-06-24 176 17:41:42:753~zt.script~Changing cameras to ignition settings
2020-06-24 176 17:41:42:754~zt.script~Starting bracketed exposures on ACME camera
2020-06-24 176 17:41:42:754~acme.image~"Received command: IntensifiedCameraGainAmplitude(gainAmplitudeCounts:3303)"
2020-06-24 176 17:41:42:754~zt.script~Changing IPSU(s) to test archive rate
2020-06-24 176 17:41:42:754~acme.image~"Received command: DataCameraExposureBracketing(enable:1,imagesPerExposure:1,exposures:3,exposureStartUsec:60000,exposureIncrementUsec:500)"
2020-06-24 176 17:41:42:755~acme.image~"Received command: AnalogCameraSetShutter(shutter:2)"
2020-06-24 176 17:41:42:791~acme.image~Analog command successful.
2020-06-24 176 17:41:42:791~acme.image~"Received command: IpsuSetArchiveRate(ipsu:1,archiveRate:2)"
2020-06-24 176 17:41:42:854~zt.script~Preheating igniter with 4.49V for 2000 Ms
2020-06-24 176 17:41:42:854~acme.image~"Received command: IpsuSetArchiveRate(ipsu:2,archiveRate:1)"
2020-06-24 176 17:41:42:855~acme.igniter~"Received command: IgnitionDetectionSetState(state:1)"
2020-06-24 176 17:41:42:855~acme.igniter~Ignition detection enabled
2020-06-24 176 17:41:42:855~acme.igniter~"Received command: IgniterPulse(power:4.49,durationMs:2500)"
2020-06-24 176 17:41:42:935~acme.ipsu~Success for IPSU 1 command IpsuEnableArchive

2020-06-24 176 17:41:42:961~acme.ipsu~Success for IPSU 2 command IpsuEnableArchive
2020-06-24 176 17:41:44:610~zt.script~Starting fuel flow at ignition flow rate
2020-06-24 176 17:41:44:670~acme.flow~"Received command: FlowSetAllSolenoidValveStates(SVA1State:0,SVA2State:0,SVA3State:0,SVA4State:0,SVA5State:0,SVA6State:0,SVA7State:0,SVA8State:0,SVA9State:1)"
2020-06-24 176 17:41:44:861~zt.script~Igniting with 4.49V for 500 Ms
2020-06-24 176 17:41:44:861~acme.igniter~"Received command: IgniterPulse(power:4.49,durationMs:500)"
2020-06-24 176 17:41:45:112~zt.script~Flow timeout in 120 sec
2020-06-24 176 17:41:45:112~acme.flow~"Received command: FlowSetAllMfcRates(MFCA1StaticMixerFlowSlm:0,MFCA2BypassFlowSlm:0,MFCA3FuelFlowSlm:0.60319)"
2020-06-24 176 17:41:45:363~zt.script~Retracting igniter
2020-06-24 176 17:41:45:363~acme.igniter~"Received command: IgnitionDetectionSetState(state:0)"
2020-06-24 176 17:41:45:363~acme.igniter~Ignition detection disabled
2020-06-24 176 17:41:45:363~acme.igniter~"Received command: IgniterExecuteSetupMove()"
2020-06-24 176 17:41:45:413~zt.script~Changing cameras to test settings
2020-06-24 176 17:41:45:413~zt.script~Waiting stabilization time #1 0 sec
2020-06-24 176 17:41:45:413~zt.script~Extinction detection enabled
2020-06-24 176 17:41:45:413~zt.script~Waiting stabilization time #2 0 sec
2020-06-24 176 17:41:45:413~zt.script~Moving fiber to test position
2020-06-24 176 17:41:45:507~acme.image~"Received command: AnalogCameraSetShutter(shutter:2)"
2020-06-24 176 17:41:45:549~acme.image~Analog command successful.
2020-06-24 176 17:41:45:549~acme.diag~"Received command: ExtinctionDetectionSetState(state:1)"
2020-06-24 176 17:41:45:550~acme.fiber~"Received command: FiberMove(position:14100,velocity:3000)"
2020-06-24 176 17:41:47:414~zt.script~Waiting constant flow time 45 sec
2020-06-24 176 17:41:59:418~zt~Script exception thrown from main script: "EXTINCTION DETECTED" line Number: "1042"
2020-06-24 176 17:41:59:418~zt.script~s-Flame test point 20176B1 ABORTED. Reason: EXTINCTION DETECTED
2020-06-24 176 17:41:59:418~zt.script~Changing cameras to extinction settings
2020-06-24 176 17:41:59:418~zt.script~Turning off gas flow
2020-06-24 176 17:41:59:418~acme.image~"Received command: AnalogCameraSetShutter(shutter:0)"
2020-06-24 176 17:41:59:455~acme.image~Analog command successful.
2020-06-24 176 17:41:59:456~acme.igniter~"Received command: IgnitionDetectionSetState(state:0)"
2020-06-24 176 17:41:59:456~acme.igniter~Ignition detection disabled
2020-06-24 176 17:41:59:456~acme.diag~"Received command: ExtinctionDetectionSetState(state:0)"
2020-06-24 176 17:41:59:457~acme.flow~"Received command: FlowCloseAll()"
2020-06-24 176 17:41:59:458~acme.fiber~"Received command: FiberHaltMove()"

2020-06-24 176 17:42:00:419~zt.script~Waiting post-flame imaging duration of 3 sec
2020-06-24 176 17:42:00:419~acme~Sent IopRunScript command to IOP.
2020-06-24 176 17:42:03:420~zt.script~Stopping IPSU(s)
2020-06-24 176 17:42:03:420~zt.script~Waiting post-flame data duration of 15 sec
2020-06-24 176 17:42:03:420~acme.image~"Received command: StopIpsus()"
2020-06-24 176 17:42:03:550~acme.ipsu~Success for IPSU 1 command IpsuTestpointStop
2020-06-24 176 17:42:03:672~acme.ipsu~Success for IPSU 2 command IpsuTestpointStop
2020-06-24 176 17:42:03:673~acme.ipsu~Success for IPSU 5 command IpsugCamerasStopTestPointArchive
2020-06-24 176 17:42:09:029~acme.ipsu~Success for IPSU 1 command IpsuTestpointClose
2020-06-24 176 17:42:16:987~zt.fcf~ -$$$$$$$ PRIORITY INVERSION:8e9d000 sendRvc 59 100
2020-06-24 176 17:42:16:988~acme.ipsu~Success for IPSU 2 command IpsuTestpointClose
2020-06-24 176 17:42:16:993~acme.ipsu~Success for IPSU 5 command IpsugTestpointClose
2020-06-24 176 17:42:18:424~zt.script~Homing igniter
2020-06-24 176 17:42:18:424~acme~"Received command: SystemCloseTestPoint()"
2020-06-24 176 17:42:18:424~zt.script~Homing fiber
2020-06-24 176 17:42:18:424~acme~List of files for test point "20176B1"
2020-06-24 176 17:42:18:424~acme~"/payload/sci/20176B1/20176B1_data.csv"
2020-06-24 176 17:42:18:424~acme~"/payload/sci/20176B1/20176B1_mfcSetpoints.csv"
2020-06-24 176 17:42:18:424~zt.script~Intensified camera to safe mode
2020-06-24 176 17:42:18:428~acme~"/payload/logs/2020-06-24_15-10_ACME.log"
2020-06-24 176 17:42:18:428~acme~"/payload/logs/2020-06-24_15-30_ACMECommands.log"
2020-06-24 176 17:42:18:428~acme.igniter~"Received command: IgniterMove(position:0)"
2020-06-24 176 17:42:18:458~acme.fiber~"Received command: FiberMove(position:0,velocity:3000)"
2020-06-24 176 17:42:18:748~acme.image~"Received command: IntensifiedCameraGate(gateType:0,pulsedGateWidthCounts:0)"
2020-06-24 176 17:42:18:925~zt.script~Setting PMT gains to zero
2020-06-24 176 17:42:18:925~acme.image~"Received command: IntensifiedCameraGainAmplitude(gainAmplitudeCounts:2048)"
2020-06-24 176 17:42:18:925~zt.script~Reporting final MFC integrated flow
2020-06-24 176 17:42:18:925~acme.image~"Received command: DataCameraExposureBracketing(enable:0,imagesPerExposure:1,exposures:3,exposureStartUsec:60000,exposureIncrementUsec:500)"
2020-06-24 176 17:42:18:926~acme.diag~"Received command: PmtSetAllGains(PMT1Gain:0,PMT2Gain:0,PMT3Gain:0)"
2020-06-24 176 17:42:18:927~acme.flow~"Received command: FlowMfcSerialCommand(serialCommand:*00g31)"
2020-06-24 176 17:42:19:127~acme.flow~Reply for command: *00g31
2020-06-24 176 17:42:19:127~acme.flow~"0
2020-06-24 176 17:42:19:127~acme.flow~>"
2020-06-24 176 17:42:19:137~zt.fcf~ -$$$$$$$ PRIORITY INVERSION:8e9d000 sendRvc 59 100
2020-06-24 176 17:42:19:937~acme.flow~"Received command: FlowMfcSerialCommand(serialCommand:*01g31)"

2020-06-24 176 17:42:20:138~acme.flow~Reply for command: *01g31
2020-06-24 176 17:42:20:138~acme.flow~"0
2020-06-24 176 17:42:20:138~acme.flow~>"
2020-06-24 176 17:42:20:938~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*02g31)"
2020-06-24 176 17:42:21:137~acme.flow~Reply for command: *02g31
2020-06-24 176 17:42:21:138~acme.flow~"0.1377819
2020-06-24 176 17:42:21:138~acme.flow~>"
2020-06-24 176 17:42:21:926~zt.script~s-Flame test point 20176B1 complete after abort
2020-06-24 176 17:42:43:880~acme.image~"Received command:
IlluminationConfigure(sideLEDBankState:1,endLEDBankState:1)"

IPSUG.log excerpt

Meanwhile, the IPSUG.log files can provide a check of the exposure time, which can be particularly important when bracketing is employed with the ACME DataCam. The corresponding excerpt for test 20176B1, in which bracketing is used, is presented here.

2020-06-24 176 17:36:20:873~ipsug~CPU operational minutes is now 49198
2020-06-24 176 17:40:40:735~ipsug.Halcon~SetExposure(integrationTimeUsecs 100000)
2020-06-24 176 17:40:54:092~ipsug~void
IpsugAvionicsController::handleTSyncCommand(QDateTime) received TIME
QDateTime("2020-06-24 17:40:54.000 UTC Qt::LocalTime")
2020-06-24 176 17:41:25:477~zt.filesys~Set new directory permissions to rwxrwxrwx (777)
for: "/sd1"
2020-06-24 176 17:41:25:477~zt.filesys~Set new directory permissions to rwxrwxrwx (777)
for: "/sd1/sci"
2020-06-24 176 17:41:25:478~zt.filesys~Set new directory permissions to rwxrwxrwx (777)
for: "/sd1/sci/20176B1"
2020-06-24 176 17:41:25:929~ipsug.image~virtual void
ImagingController::handleCommand(XGenericMessage*) bCmdComplete camera 3
2020-06-24 176 17:41:25:929~ipsug.image~virtual void
ImagingController::handleCommand(XGenericMessage*) bCmdComplete camera 4
2020-06-24 176 17:41:25:929~ipsug.FGrabber~void HalconFrameGrabber::captureImages()
"ACME" start of capture
2020-06-24 176 17:41:25:930~ipsug.FGrabber~void HalconFrameGrabber::captureImages()
"INTNS" start of capture
2020-06-24 176 17:41:42:882~ipsug.Halcon~setExposureBracketing(1 , 3 , 60000 , 500)
2020-06-24 176 17:41:42:882~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:42:885~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:42:985~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:43:284~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:44:783~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:44:842~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:45:144~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:46:643~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:46:701~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:47:003~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)

2020-06-24 176 17:41:48:502~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:48:565~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:48:863~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:50:364~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:50:424~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:50:722~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:52:222~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:52:284~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:52:583~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:54:082~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:54:141~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:54:443~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:55:944~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:56:004~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:56:303~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:57:804~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:57:865~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:41:58:162~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:41:59:664~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:41:59:721~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:42:00:022~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:42:01:523~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:42:01:584~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:42:01:883~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:42:03:383~ipsug.Halcon~SetExposure(integrationTimeUsecs 300000)
2020-06-24 176 17:42:03:445~ipsug.Halcon~SetExposure(integrationTimeUsecs 1500000)
2020-06-24 176 17:42:03:623~ipsug.FGrabber~"INTNS" saving meta data after archiving
2020-06-24 176 17:42:03:685~ipsug.FGrabber~"INTNS" end of capture in captureImages
2020-06-24 176 17:42:03:737~ipsug.FGrabber~"ACME" saving meta data after archiving
2020-06-24 176 17:42:03:738~ipsug.Halcon~SetExposure(integrationTimeUsecs 60000)
2020-06-24 176 17:42:03:795~ipsug.FGrabber~"ACME" end of capture in captureImages
2020-06-24 176 17:42:19:136~ipsug.Halcon~setExposureBracketing(0 , 0 , 0 , 0)

Appendix A
Nomenclature

This document's key abbreviations are listed here.

ACME	Advanced Combustion via Microgravity Experiments (project including s- Flame)
AOI	Area of Interest (e.g., for image acquisition of just a portion of the camera FOV)
ASCII	American Standard Code for Information Interchange
BRE	Burning Rate Emulator (another ACME experiment)
CCD	Charge-Coupled Device (such as a camera sensor)
CIR	Combustion Integrated Rack (ISS facility in which ACME experiments are conducted)
CFI	Cool Flames Investigation (a droplet combustion experiment preceding ACME in the CIR)
CLD Flame	Coflow Laminar Diffusion Flame (another ACME experiment)
CSV	comma-separated values
DataCam	Data Camera (for ACME experiments, characterized for pyrometry, etc.)
E-FIELD Flames	Electric-Field Effects on Laminar Diffusion Flames
Fiji	Fiji Is Just ImageJ (a version of ImageJ preloaded with many plug-in features)
FOMA	Fuel Oxidizer Management Assembly (CIR's gas management subsystem)
FOV	Field of View
fps	frames per second
GCIP	Gas Chromatograph Instrumentation Package
GIPOC	Glenn ISS Payload Operations Center
GMT	Greenwich Mean Time
HiBMs	High Bit-depth Multi-spectral camera (of the CIR facility)
HOBJ	Halcon object (binary uncompressed format; used for the DataCam)
ImageJ	Image Processing and Analysis in Java (open-source image processing software)
IPSU	Image Processing and Storage Unit (CIR subsystem)
IPSU-G	IPSU for Gigabit ethernet cameras
ISS	International Space Station
LLL-UV	Low Light Level Ultraviolet camera (of the CIR facility)
MFC	Mass Flow Controller
NTSC	National Television System Committee
OMA2	Optical Multi-channel Analyzer 2 (open-source software for image processing and display)
ops	operations
PaRIS	Passive Rack Isolation System
PIMS	Principal Investigator Microgravity Services
PMT	Photomultiplier Tube (where ACME is equipped with three PMT modules)
PT	pressure transducer
psia	pounds per square inch
s-Flame	Structure and Response of Spherical Diffusion Flame (another ACME experiment)
SAMS	Space Acceleration Measurement System
slpm	standard liters per minute (a measure of the mass or molar flow rate of a gas)
TFP	Thin-Filament Pyrometry (a technique for making gas-phase temperature measurements)
TIFF or TIF	Tagged Image File Format (used by the CFI, HiBMs, and LLL-UV cameras)
TM	thermistor
TS	Transport Stream (a video format)

Appendix B
Installation of QtOMA2 on a PC

These instructions were provided by Prof. Marshall Long (Yale U.).

These instructions assume you don't already have Qt installed. If you do, skip to Part B.

Part A:

Download the opensource Qt from https://www.qt.io/download.

- ➔ Go open source
- ➔ Download the Qt Online Installer
- ➔ Download
- ➔ Save File

In the downloads folder, open qt-unified-windows-x86-x.x.x-online

- ➔ Sign up for a Qt account or login
- ➔ Accept Open Source obligations
- ➔ Contribute to Qt Development (on not)
- ➔ Accept default install location
- ➔ Install the components as shown in the figure on the next page
- ➔ Agree to terms
- ➔ Add shortcuts
- ➔ Install

Part B:

Download the latest version of QtOma2 from https://github.com/mblong/qtoma2.

- ➔ Code → Download ZIP

Expand the zip file.
In the qtoma2-master folder, open QtOma.pro (the Qt Project file)

The first time this opens in Qt Creator, it will ask for the project to be configured. Just choose the defaults.

- ➔ Configure Project
- ➔ In the lower left corner, click on the green arrow to run QtOma2.

Selection of Qt (not OMA2) components to install in part A

